MyQuest™

HAO,JENNIFER PEI RU

DOB: 02/15/1994
Sex: F
Phone: (917) 743-1266
Patient ID: 6101106

Age: 32
Fasting: Y

Specimen: SA671940W
Requisition: 7701199
Report Status: FINAL / SEE REPORT

Collected: 02/26/2026 09:30
Received: 02/27/2026 07:09
Reported: 03/07/2026 17:02

Client #: 41141141
TAYLOR,NADIA
AIQ REPORTED
CUSTOMER SOLUTIONS
8401 FALLBROOK AVE
WEST HILLS, CA 91304-3226
Phone: (818) 737-6000

⚠ METHYLMALONIC ACID AND HOMOCYSTEINE

Analyte	Value	
⚠ **METHYLMALONIC ACID**	53 L	Reference Range: 55-335 nmol/L

Serum methylmalonic acid (MMA) levels are used to diagnose and monitor several rare inborn errors of metabolism, including methylmalonic aciduria. The enzymatic conversion of MMA to succinic acid requires vitamin B12 (adenosyl-cobalamin) as a cofactor. Serum MMA levels are also used for assessing functional vitamin B12 deficiency. Vitamin B12 is essential for fetal neurodevelopment, particularly early in pregnancy. Undiagnosed maternal vitamin B12 deficiency may be associated with adverse fetal/neonatal outcomes, such as neural tube defects and intrauterine growth restriction.

Quest Diagnostics utilized Multi-Modal Decomposition (MMD) analysis to establish first and second trimester-specific MMA reference intervals in pregnancy, as given below:
MMA, First trimester (<13 wks gestation): 58-167 nmol/L
MMA, Second trimester (13-23 wks gestation):
63-241 nmol/L

This test was developed and its analytical performance characteristics have been determined by Quest Diagnostics. It has not been cleared or approved by the FDA. This assay has been validated pursuant to the CLIA regulations and is used for clinical purposes.

Analyte	Value	
HOMOCYSTEINE	6.1	Reference Range: <=11.0 umol/L

Homocysteine is increased by functional deficiency of folate or vitamin B12. Testing for methylmalonic acid differentiates between these deficiencies. Other causes of increased homocysteine include renal failure, folate antagonists such as methotrexate and phenytoin, and exposure to nitrous oxide.
Selhub J, et al. Ann Intern Med. 1999;131(5):331-9.

⚠ CHOLESTEROL, TOTAL

Analyte	Value	
⚠ **CHOLESTEROL, TOTAL**	276 H	Reference Range: <200 mg/dL

⚠ DIRECT LDL

Analyte	Value

⚠ DIRECT LDL	198 H	Reference Range: <100 mg/dL

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Greatly elevated Triglycerides values (>1200 mg/dL)
interfere with the dLDL assay.
LDL-C levels > or = 190 mg/dL may indicate familial
hypercholesterolemia (FH). Clinical assessment and
measurement of blood lipid levels should be
considered for all first degree relatives of patients
with an FH diagnosis. LDL Cholesterol (LDL-C)
levels > or = 300 mg/dL may indicate homozygous
familial hypercholesterolemia (HoFH). Untreated,
these extremely high LDL-C levels can result in
premature CV events and mortality. Patients should
be identified early and provided appropriate
interventions to reduce the cumulative LDL-C
burden from birth.

For questions about testing for familial
hypercholesterolemia, please call Quest
Genomics Client Services at 1.866.GENE.INFO.

Jacobson T, et al. J National Lipid Association
Recommendations for Patient-Centered
Management of Dyslipidemia: Part 1 Journal of
Clinical Lipidology 2015;9(2), 129-169.
Cuchel, M. et al. (2014). Homozygous familial
hypercholesterolaemia: new insights and guidance
for clinicians to improve detection and clinical
management. European Heart Journal, 35(32),
2146-2157.

Desirable range <100 mg/dL for primary prevention;
<70 mg/dL for patients with CHD or diabetic patients
with > or = 2 CHD risk factors.
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⚠ PREGNENOLONE, LC/MS

Analyte	Value	
⚠ PREGNENOLONE, LC/MS	8 L	Reference Range: 22-237 ng/dL See Note 1

⚠ OMEGACHECK®

Analyte	Value	
EPA+DPA+DHA	9.2	Reference Range: >5.4 % by wt

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This test was developed and its analytical performance
characteristics have been determined by Quest Diagnostics
Cardiometabolic Center of Excellence at Cleveland HeartLab.
It has not been cleared or approved by the U.S. Food and
Drug Administration. This assay has been validated pursuant
to the CLIA regulations and is used for clinical purposes.
Increasing blood levels of long-chain n-3 fatty acids are
associated with a lower risk of sudden cardiac death (1).
Based on the top (75th percentile) and bottom (25th
percentile) quartiles of the CHL reference population, the
following relative risk categories were established for
OmegaCheck:  A cut-off of >=5.5% by wt defines a population
at optimal relative risk, 3.8-5.4% by wt defines a
population at moderate relative risk, and <=3.7% by wt
defines a population at high relative risk of sudden
cardiac death. The totality of the scientific evidence
demonstrates that when consumption of fish oils is limited
to 3 g/day or less of EPA and DHA, there is no significant
risk for increased bleeding time beyond the normal range. A
daily dosage of 1 gram of EPA and DHA lowers the
circulating triglycerides by about 7-10% within 2 to 3
weeks. (Reference: 1-Albert et al. NEJM. 2002; 346:
1113-1118).
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ARACHIDONIC ACID/EPA RATIO	4.5	Reference Range: 3.7-40.7
OMEGA-6/OMEGA-3 RATIO	4.3	Reference Range: 3.7-14.4
OMEGA-3 TOTAL	9.2	% by wt
⚠ EPA	2.7 H	Reference Range: 0.2-2.3 % by wt
DPA	1.3	Reference Range: 0.8-1.8 % by wt
⚠ DHA	5.3 H	Reference Range: 1.4-5.1 % by wt

OMEGA-6 TOTAL Cleveland HeartLab measures a number of omega-6 fatty acids with AA and LA being the two most abundant forms reported.	39.9	% by wt
ARACHIDONIC ACID	12.0	Reference Range: 8.6-15.6 % by wt
LINOLEIC ACID	25.6	Reference Range: 18.6-29.5 % by wt

CELIAC DISEASE COMPREHENSIVE PANEL

Analyte	Value	
INTERPRETATION No serological evidence of celiac disease. tTG IgA may normalize in individuals with celiac disease who maintain a gluten-free diet. Consider HLA DQ2 and DQ8 testing to rule out celiac disease. Celiac disease is extremely rare in the absence of DQ2 or DQ8.	SEE NOTE	
TISSUE TRANSGLUTAMINASE AB, IGA Units Value Interpretation ----------- -------------- <15.0 Antibody not detected > or = 15.0 Antibody detected	<1.0	U/mL
IMMUNOGLOBULIN A	270	Reference Range: 47-310 mg/dL

HOUSE ACCOUNT TRACKING

Analyte	Value
TRACKING HOUSE ACCOUNT We were unable to identify an account number for the order submitted. If you do not have a Quest Diagnostics account number or if your account information needs to be updated please call 1-866-MYQUEST (866-697-8378) for assistance. To prevent delays in testing and processing of your orders please provide the following information for this order and with every additional order submitted: Quest account number and account name Client address Client phone and fax number NPI number of ordering physician along with the physician name.	

TEST AUTHORIZATION

Analyte	Value	
TEST NAME:	OMEGACHECK(R)	
TEST CODE:	92701SBX	
CLIENT CONTACT:	NADIA TAYLOR	
REPORT ALWAYS MESSAGE SIGNATURE		See Note 2
COMMENT		See Note 3

TEST AUTHORIZATION 2

Analyte	Value	
TEST NAME:	LYME DISEASE ANTIBODIES	
TEST CODE:	8593RXE	
CLIENT CONTACT:	NADIA TAYLOR	
REPORT ALWAYS MESSAGE SIGNATURE		See Note 2

COMMENT		See Note 3

COPPER

Analyte	Value	
COPPER	Test Not Performed	mcg/dL See Note 4

ZINC

Analyte	Value	
ZINC	Test Not Performed	mcg/dL See Note 4

IRON, TIBC AND FERRITIN PANEL

Analyte	Value	
FERRITIN	46	Reference Range: 16-154 ng/mL

IRON AND TOTAL IRON BINDING CAPACITY

Analyte	Value	
IRON, TOTAL	91	Reference Range: 40-190 mcg/dL
IRON BINDING CAPACITY	313	Reference Range: 250-450 mcg/dL (calc)
% SATURATION	29	Reference Range: 16-45 % (calc)

HDL CHOLESTEROL

Analyte	Value	
HDL CHOLESTEROL	55	Reference Range: > OR = 50 mg/dL

HS CRP

Analyte	Value	
HS CRP	2.1	mg/L

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Reference Range
Optimal <1.0
Jellinger PS et al. Endocr Pract.2017;23(Suppl 2):1-87.
For ages >17 Years:
hs-CRP mg/L  Risk According to AHA/CDC Guidelines
<1.0         Lower relative cardiovascular risk.
1.0-3.0      Average relative cardiovascular risk.
3.1-10.0     Higher relative cardiovascular risk.
             Consider retesting in 1 to 2 weeks to
             exclude a benign transient elevation
             in the baseline CRP value secondary
             to infection or inflammation.
>10.0        Persistent elevation, upon retesting,
             may be associated with infection and
             inflammation.

Pearson TA, Mensah GA, Alexander RW, et al. Markers
of inflammation and cardiovascular disease:
application to clinical and public health practice:
A statement for healthcare professionals from the
Centers for Disease Control and Prevention and the
American Heart Association. Circulation 2003; 107(3):
499-511.
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APOLIPOPROTEIN A1

Analyte	Value

APOLIPOPROTEIN A1	140	mg/dL

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Reference Range    > or = 125

Risk Category:
Optimal         > or = 125
High            <125

Cardiovascular event risk category
cut points (optimal, high) are based
on the AMORIS study, Walldius G et
al. J Intern Med. 2004;255:188-205.
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COMPREHENSIVE METABOLIC PANEL

Analyte	Value	
GLUCOSE Fasting reference interval	89	Reference Range: 65-99 mg/dL
UREA NITROGEN (BUN)	11	Reference Range: 7-25 mg/dL
CREATININE	0.68	Reference Range: 0.50-0.97 mg/dL
EGFR	119	Reference Range: > OR = 60 mL/min/1.73m2
BUN/CREATININE RATIO	16	Reference Range: 6-22 (calc)
SODIUM	138	Reference Range: 135-146 mmol/L
POTASSIUM	4.3	Reference Range: 3.5-5.3 mmol/L
CHLORIDE	104	Reference Range: 98-110 mmol/L
CARBON DIOXIDE	27	Reference Range: 20-32 mmol/L
CALCIUM	9.5	Reference Range: 8.6-10.2 mg/dL
PROTEIN, TOTAL	7.4	Reference Range: 6.1-8.1 g/dL
ALBUMIN	4.5	Reference Range: 3.6-5.1 g/dL
GLOBULIN	2.9	Reference Range: 1.9-3.7 g/dL (calc)
ALBUMIN/GLOBULIN RATIO	1.6	Reference Range: 1.0-2.5 (calc)
BILIRUBIN, TOTAL	0.4	Reference Range: 0.2-1.2 mg/dL
ALKALINE PHOSPHATASE	45	Reference Range: 31-125 U/L
AST	17	Reference Range: 10-30 U/L
ALT	18	Reference Range: 6-29 U/L

HEMOGLOBIN A1c WITH eAG

Analyte	Value	
HEMOGLOBIN A1c	5.5	Reference Range: <5.7 %

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For the purpose of screening for the presence of
diabetes:

<5.7%       Consistent with the absence of diabetes
5.7-6.4%    Consistent with increased risk for diabetes
            (prediabetes)
> or =6.5%  Consistent with diabetes

This assay result is consistent with a decreased risk
of diabetes.

Currently, no consensus exists regarding use of
hemoglobin A1c for diagnosis of diabetes in children.

According to American Diabetes Association (ADA)
guidelines, hemoglobin A1c <7.0% represents optimal
control in non-pregnant diabetic patients. Different
metrics may apply to specific patient populations.
Standards of Medical Care in Diabetes(ADA).
```

eAG (mg/dL)	111	mg/dL
eAG (mmol/L)	6.2	mmol/L

MAGNESIUM

Analyte	Value	
MAGNESIUM	2.2	Reference Range: 1.5-2.5 mg/dL

T4 (THYROXINE), TOTAL

Analyte	Value	
T4 (THYROXINE), TOTAL	7.0	Reference Range: 5.1-11.9 mcg/dL

T3 REVERSE, LC/MS/MS

Analyte	Value	
T3 REVERSE, LC/MS/MS	12	Reference Range: 8-25 ng/dL See Note 1

SED RATE BY MODIFIED WESTERGREN

Analyte	Value	
SED RATE BY MODIFIED WESTERGREN	14	Reference Range: < OR = 20 mm/h

VITAMIN B12 (REFL)

Analyte	Value	
VITAMIN B12	683	Reference Range: 200-1100 pg/mL

ANA SCREEN, IFA, W/REFL TITER AND PATTERN

Analyte	Value	
ANA SCREEN, IFA	NEGATIVE	Reference Range: NEGATIVE

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ANA IFA is a first line screen for detecting the presence of up to approximately 150 autoantibodies in
various autoimmune diseases. A negative ANA IFA result suggests an ANA-associated autoimmune disease is not
present at this time, but is not definitive. If there is high clinical suspicion for Sjogren's syndrome,
testing for anti-SS-A/Ro antibody should be considered. Anti-Jo-1 antibody should be considered for
clinically suspected inflammatory myopathies.

AC-0: Negative

International Consensus on ANA Patterns (https://doi.org/10.1515/cclm-2018-0052)

For additional information, please refer to http://education.QuestDiagnostics.com/faq/FAQ177 (This link is
being provided for informational/ educational purposes only.)
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CERULOPLASMIN

Analyte	Value	
CERULOPLASMIN	23	Reference Range: 14-48 mg/dL

RHEUMATOID FACTOR

Analyte	Value	
RHEUMATOID FACTOR	<10	Reference Range: <14 IU/mL

INSULIN

Analyte	Value	
INSULIN	13.4	Reference Range: < OR = 18.4 uIU/mL

```
Risk:
Optimal < or = 18.4
Moderate NA
High >18.4

Adult cardiovascular event risk category cut points
(optimal, moderate, high) are based on Insulin Reference
interval studies performed at Quest Diagnostics in 2022.
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CORTISOL, TOTAL, LC/MS

Analyte	Value	
CORTISOL, TOTAL, LC/MS	7.1	mcg/dL

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Adult Reference Ranges for Cortisol, Total:

   8-10 AM  4.6-20.6 mcg/dL
   4-6  PM  1.8-13.6 mcg/dL

   Cortisol Response to ACTH
   Peak >20.0 mcg/dL
   Peak >16.0 mcg/dL after IM injection

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characteristics have been determined by Quest Diagnostics.
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has been validated pursuant to the CLIA regulations and is
used for clinical purposes.
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LYME DISEASE ANTIBODIES (IGG,IGM), IMMUNOBLOT

Analyte	Value
LYME DISEASE AB(IGG),BLOT	NEGATIVE
18 KD (IGG) BAND	NON-REACTIVE
23 KD (IGG) BAND	NON-REACTIVE
28 KD (IGG) BAND	NON-REACTIVE
30 KD (IGG) BAND	NON-REACTIVE
39 KD (IGG) BAND	NON-REACTIVE
41 KD (IGG) BAND	REACTIVE
45 KD (IGG) BAND	NON-REACTIVE
58 KD (IGG) BAND	REACTIVE
66 KD (IGG) BAND	NON-REACTIVE
93 KD (IGG) BAND	NON-REACTIVE
LYME DISEASE AB(IGM),BLOT	NEGATIVE
23 KD (IGM) BAND	NON-REACTIVE
39 KD (IGM) BAND	NON-REACTIVE

41 KD (IGM) BAND	**NON-REACTIVE**

REFERENCE RANGE: NEGATIVE

Lyme immunoblot testing should only be performed on samples from patients who have had a Positive or Equivocal result in a screening assay.

As per CDC criteria, a Lyme disease IgG immunoblot must show reactivity to at least 5 of 10 specific borrelial proteins to be considered positive; similarly, a positive Lyme disease IgM immunoblot requires reactivity to 2 of 3 specific borrelial proteins. Although considered negative, IgG reactivity to fewer specific borrelial proteins or IgM reactivity to only 1 protein may indicate recent B. burgdorferi infection and warrant testing of a later sample. A positive IgM but negative IgG result obtained more than a month after onset of symptoms likely represents a false-positive IgM result rather than acute Lyme disease. In rare instances, Lyme disease immunoblot reactivity may represent antibodies induced by exposure to other spirochetes.

Note 1	This test was developed and its analytical performance characteristics have been determined by Quest Diagnostics. It has not been cleared or approved by the FDA. This assay has been validated pursuant to the CLIA regulations and is used for clinical purposes.
Note 2	The laboratory testing on this patient was verbally requested or confirmed by the ordering physician or his or her authorized representative after contact with an employee of Quest Diagnostics. Federal regulations require that we maintain on file written authorization for all laboratory testing. Accordingly we are asking that the ordering physician or his or her authorized representative sign a copy of this report and promptly return it to the client service representative. Signature:____
Note 3	Please fax this signed form to 1-813-605-7586. Please do not attempt to return this document by other methods. Documents will not be viewed by a representative. Please do not use this fax number for other service requests.
Note 4	TEST NOT PERFORMED No suitable specimen received. Please review the test requirements at testdirectory.questdiagnostics.com

Copies Sent To

NADIA TAYLOR, 554 CLAYTON ST #170008, SAN FRANCISCO,CA, 94117

Performing Sites

AMD Quest Diagnostics/Nichols Chantilly-Chantilly VA, 14225 Newbrook Dr, Chantilly, VA 20151-2228 Laboratory Director: Patrick W Mason M.D.,PhD
EN Quest Diagnostics-West Hills, 8401 Fallbrook Ave, West Hills, CA 91304-3226 Laboratory Director: Thomas J McDonald
EZ Quest Diagnostics/Nichols SJC-San Juan Capistrano,, 33608 Ortega Hwy, San Juan Capistrano, CA 92675-2042 Laboratory Director: Irina Maramica MD,PhD,MBA
UL Quest Diagnostics-Sacramento - Northgate, 3714 Northgate Blvd, Sacramento, CA 95834-1617 Laboratory Director: Lorne L Holland
Z4M Cleveland HeartLab Inc.-Cleveland HeartLab Inc., 6701 Carnegie Ave, Suite 500, Cleveland, OH 44103-4638 Laboratory Director: Mohammad Q Ansari

Key

Priority Out of Range   Out of Range